

Course Name: General College Chemistry 1

Course Description

Students who take this course learn the atomic structure and chemical properties of the elements according to their position in the Periodic table. This course enables you to use standard chemical names and symbols to represent elements, isotopes, ions, compounds, and chemical reactions. By taking this course you successfully learn to classify patterns in bonding, molecular geometry and chemical reaction. You understand the concepts of molar and its importance in the quantitative analysis of chemical reactions and make calculations easily. And you learn the laws that explain the behavior of gases in detail and successfully apply them.

Outcomes and Objectives and Evaluation Tools:

A. Outcome - by the successful completion of this course, students will demonstrate the ability to: Explain atomic structure and chemical properties of the elements based in their position in the periodic table.

A1. Define the atom

A2. Identify the names & symbols of the elements

A3. Identify the elements & compounds

A4. Calculate the atomic mass number of particles and average atomic mass number of isotopes.

A5. Explain how to use the periodic table to obtain information.

A6. Identify the composition of the periodic table.

The student will demonstrate competence:

by preparing and giving a presentation by 3-4 students of group that explains atomic structure and chemical properties of the elements based in their position in the periodic table. (%20)

Performance will be successful when: These are the criteria.

- The presentation includes a least one slide for each subtitle in the chapter

- Each group member will take in presentation 5 min
- Presentation purpose is clear
- Presentation conclusion includes a summary of the main points

B. Outcome - by the successful completion of this course, students will demonstrate the ability to: Use standard chemical names and symbols to represent elements, isotopes, ions, compounds, and chemical reactions.

- B1.** Explain what a chemical formula represents
- B2.** Explain several types of compounds
- B3.** Explain how to write formula of compounds
- B4.** Explain how to name compounds
- B5.** Identify all components in a chemical equation
- B6.** Identify the types of chemical reactions

The student will demonstrate competence:

by taking an exam that lasts two ours and includes yes-no, multiple-choice and short-essay questions. (%20)

Performance will be successful when: These are the criteria.

- Explain several types of compounds
- Explain how to write formula of compounds
- Explain how to name ionic compounds
- Explain how to name covalent compounds
- Balance chemical equations
- Identify the types of chemical reactions

C. Outcome - by the successful completion of this course, students will demonstrate the ability to: Classify patterns in bonding, molecular geometry

- C1.** Define chemical bond
- C2.** Define ionic and covalently bonding structures
- C3.** Explain polarity of chemical bonds and molecular structures.
- C4.** Explain the bonding capacity of all second-row elements.

The student will demonstrate competence:

by preparing a written response to a case study on patterns in bonding, molecular geometry, and chemical reaction (%20)

the essay format:

- Cover page includes student name, instructor name, title of the case and date.
- Maximum 5 pages
- Double space
- Times New Roman
- Font size 12
- References

Performance will be successful when: These are the criteria.

- Case study response shows definition of chemical bond
- Case study response explains why chemical bonding occurs
- Case study response identifies the bonding in nonmetals
- Case study response explains the formation of covalent bonding
- Case study response explains polarity of chemical bonds and molecular and bonding structure of

all second-row elements.

- Case study response is well organized
- Case study response uses correct grammar, punctuation, and spelling

D. Outcome - by the successful completion of this course, students will demonstrate the ability to: Apply the molar concepts to the quantitative analysis of chemical reactions

D1. Define Avogadro's number and molar mass

D2. Make the main calculations about mole number

D3. Solve the problems including stoichiometric calculations

D4. Make calculations about limiting reagent problems

D5. Calculate percent yield by mass and volume

The student will demonstrate competence:

by taking an exam that lasts two ours and includes yes-no, multiple-choice and short-essay questions (%20)

Performance will be successful when: These are the criteria.

- Define Avogadro's number and molar mass
- Make the main calculations about mole number
- Solve problems including stoichiometric calculations
- Make calculations about limiting reagent problems
- Calculate percent yield
- Solve mixture problems

E. Outcome - by the successful completion of this course, students will demonstrate the ability to: Explain the laws that explain the behavior of gases

E1. Discuss gas pressure in terms of its cause, and its relationship to the temperature, volume, and number of moles of the gas.

E2. Explain gas laws (Boyles, Charles, Avogadro)

E3. Explain the ideal gas equations ($PV=nRT$).

The student will demonstrate competence:

by creating a web site that teaches the laws that explain the behavior of gases. (%20)

Performance will be successful when: These are the criteria.

- Web project includes all gas laws
- The web site includes one page of each gas law
- The site includes at least one animation that shows any gas law
- The site includes one colorful picture that shows the formula of gas laws
- Web page buttons and navigation bars are easy to understand and use
- Web page text is readable, legible, and attractive